L4.2b Washers

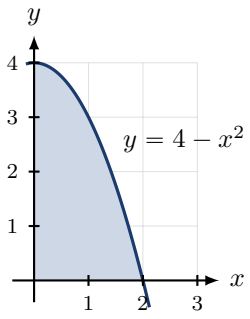
Recall: volumes by slicing

Name _____ **Solo – 9 minutes**

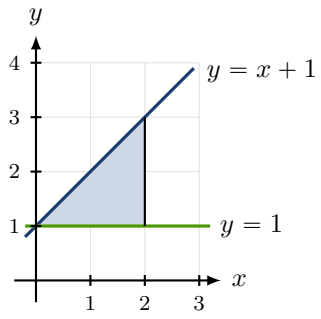
1. The region under $y = \sqrt{x+1}$ on $[0, 3]$ is rotated about the x-axis. Write the radius $r(x)$ and the chunk $dV = A(x)\,dx$, and find the volume of the solid.

2. Each region is rotated about the x-axis. Does the region touch the axis? Sketch the slice of the solid at x.

(a)



(b)



3. The region between $y = x + 1$ and $y = 1$ on $[0, 2]$ is rotated about the x-axis. Write the area of the slice at x.